



Basic Concepts in Minimal Intervention Dentistry

Learning Outcomes



- State the Basic Principles of minimal intervention dentistry.
- Compare different types of caries risk assessment
- List the different methods of minimal intervention restorative procedure.

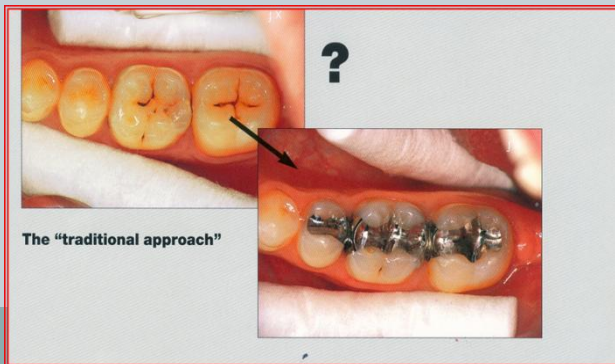
INTRODUCTION

BLACK'S TEACHINGS

- highly formalised cavity designs; precise size and geometry
 - ➔ weak, non-adhesive materials
- 'extension for prevention'

WHAT'S CHANGED?

- enhanced understanding of the carious process
 - ➔ an infectious disease
 - ➔ demineralisation/remineralisation cycle
- recognition of the rôle of fluoride
- development of adhesive materials
 - ➔ glass-ionomer cement
 - ➔ resin-based materials



OPERATIVE DENTISTRY
TWENTIETH CENTURY (GV BLACK)
Extension for prevention'

TWENTY-FIRST CENTURY
'Prevention of extension'



What is Minimal intervention dentistry



- Minimal intervention dentistry (MID) can be defined as a philosophy of professional care concerned with the **first occurrence, earliest detection and earliest possible cure of disease on microscopic (molecular) levels, followed by minimally invasive** and patient-friendly treatment to repair irreversible damage caused by such disease.

Tyas MJ, Anusavice KJ, Frencken JE, Mount GJ. Minimal intervention dentistry—a review. FDI Commission Project 1-97. Int Dent J 2000;50:1–12.



- Minimum intervention dentistry (MID) is the modern medical approach to the management of caries, utilizing caries risk assessment and focusing on the early prevention and interception of disease
- The goal of Minimal Invasive Dentistry (MID) is to stop the disease process and then to restore back the lost tooth structure and function along with maximizing the healing potential of the tooth.

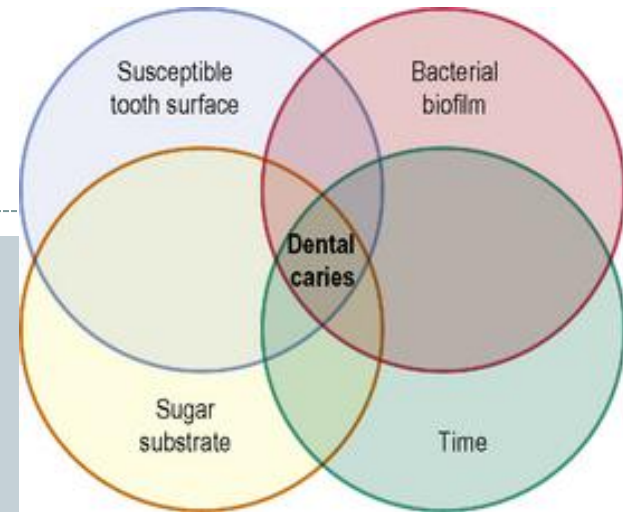
Int Dent J. 2012 October ; 62(5): 223–243

Basic principles of Minimal Intervention Dentistry



- A recent policy document produced for the World Dental Federation suggested that there are four basic principles that must be applied to fulfill the description of minimal intervention dentistry :
 - I. Control the disease through reduction of cariogenic flora and early caries detection
 - II. Remineralise early carious lesions
 - III. Perform minimal intervention restorative procedures if required
 - IV. Repair rather than replace defective restorations

Dental caries



- Dental caries is a complex process of cyclical enamel demineralization and remineralisation.
- It has been known for decades that Dental caries is a infectious disease caused by dental plaque, an oral biofilm and by exposure to fermentable carbohydrates.

REVISED CLASSIFICATION OF CARIOUS LESIONS BY SITE AND SIZE



At the FDI World Dental Congress held at Kuala Lumpur, Malaysia
in September 2001,

Dr Graham Mount said that,

“ For the first time in almost a hundred years we are now being
released from the cavity classification proposed by Dr G.V.Black...
He classified cavities, I would much rather classify caries...”

- The original approach to the treatment of caries was purely surgical.
- Cavity designs were classified and standardized and sound natural tooth structure was sacrificed in the name of geometric perfection to accommodate the shortcomings of the restorative material.

G.J.Mount Classification of carious lesions

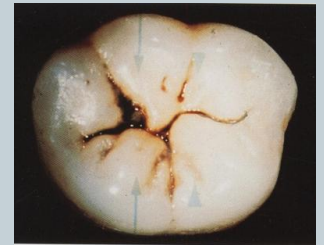


LOCATION	CLASSIFICATION			
	1=Minimal	2=Moderate	3=Advanced	4=Extensive
Site 1: Pits & Fissures	1.1	1.2	1.3	1.4
Site 2: Proximal Surfaces	2.1	2.2	2.3	2.4
Site 3: Cervical Surfaces	3.1	3.2	3.2	3.4





- Firstly, lesions are classified according to their location:
 - Site 1: Pits and fissures (occlusal and other smooth tooth surfaces)
 - Site 2: Contact area between two teeth
 - Site 3: Cervical area in contact with gingival tissues





- **Secondly**, the new classification identifies carious lesions according to various sizes:



- Size 1: Minimal change which can be managed by healing through remineralization

- Size 2: Moderate cavity not extended to cusps



- Size 3: Advanced cavity, with undermined cusp which needs protection from occlusal load



- Size 4: Extensive cavity, with at least one missing cusp or incisal edge



I. Early diagnosis of caries



- The traditional technique used for detecting carious lesions are dental explorer and the conventional X-ray.
- Radiography is reliable for detecting carious lesions in proximal tooth surfaces, but considered unreliable in occlusal surfaces, particularly for diagnosing early(initial) carious lesions in enamel.



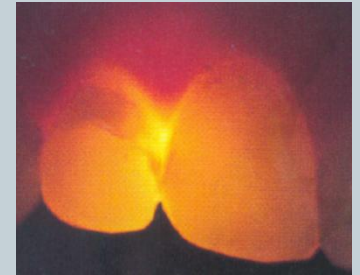
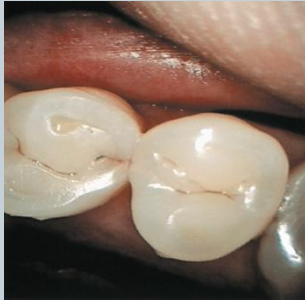
Shallow caries



Deep caries

Newer techniques for early diagnosis of Caries lesions

1. FibreOptic Trans-Illumination (FOTI)



Principle:

- It works with the principle of light scattering . The light scatters more when there is caries than in non carious tissue .
- It is observed as dark shadow against light back ground.

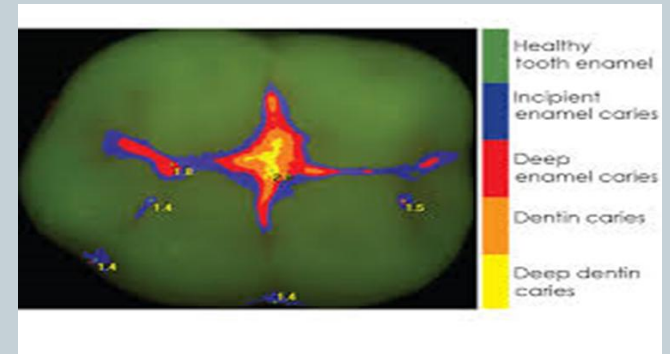


(Spectra Camera)



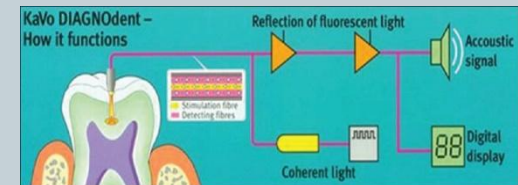
2. Quantitative light-induced fluorescence (QLF)

- Reduction in the fluorescence of demineralised enamel
- Measures actual loss of mineral and evaluates changes over time



3. DIAGNOdent

- *It is a battery-operated quantitative diode laser machine*
- *Decay emits flouoroscent light and taken as reading by sensor*



4. Electrical impedance (CarieScan)

Based on the principle of Changes in electrical impedance between sound enamel and demineralized enamel



Caries risk assessment



- Caries risk is defined as ‘the probability of future caries disease development’.

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Caries Risk Assessment Form (Age >6)

Patient Name: _____

Birth Date: _____ Birth: _____

Age: _____

	Low Risk	Moderate Risk	High Risk
Contributing conditions			
Check or Circle the conditions that apply			
1. Frequent Exposure to drinking water, supplements, professional products, toothpaste	☐ No	☐ No	☐ No
2. Sugary foods or liquids (including carbonated non-saturated soft drinks, energy drinks, and soft drinks)	☐ No	☐ No	☐ No
3. Caries Experience of Patient, Caries Index and/or DMFT (the patient's age is 6-12)	☐ No	☐ No	☐ No
4. Systemic medical conditions, chronic or acute, including immune deficiency disorders	☐ No	☐ No	☐ No
General Health Conditions			
Check or Circle the conditions that apply			
1. Special Health Care Needs (developmental, physical, mental, or behavioral) that require special care or attention	☐ No	☐ No	☐ No
2. Chronic Medication Therapy	☐ No	☐ No	☐ No
3. Eating Disorders	☐ No	☐ No	☐ No
4. Medication that reduces salivary flow	☐ No	☐ No	☐ No
5. Deep Mouth Wound	☐ No	☐ No	☐ No
Clinical Conditions			
Check or Circle the conditions that apply			
1. Caries or other dental conditions that require treatment or attention	☐ No	☐ No	☐ No
2. Periodontal Disease	☐ No	☐ No	☐ No
3. Recent Missing Root Surfaces in past 12 months	☐ No	☐ No	☐ No
4. Multiple Fractures	☐ No	☐ No	☐ No
5. Structural Tooth Damage from carious or traumatic injury	☐ No	☐ No	☐ No
6. Malocclusion or Malocclusion - 1 or more	☐ No	☐ No	☐ No
7. Periodontal Health Index	☐ No	☐ No	☐ No
8. Restorations with overhangs, under margins, open margins, or other defects	☐ No	☐ No	☐ No
9. Restorations with defective margins or other defects	☐ No	☐ No	☐ No
10. Secondary Mouth Infection	☐ No	☐ No	☐ No
Overall assessment of dental caries risk: ☐ Low ☐ Moderate ☐ High			



- Patients can be assessed and categorized into 3 groups :

Low Risk

1. No caries in last few years
2. Sealed pit and fissure
3. Good oral hygiene
4. Appropriate fluoride use
5. Regular dental visits

Moderate Risk

1. One carious lesion in last 3 year
2. Deep pits and fissures
3. Fair oral hygiene
4. White spots/ interproximal radiolucencies
5. Inadequate fluoride exposure
6. Irregular dental visits
7. Orthodontic treatment



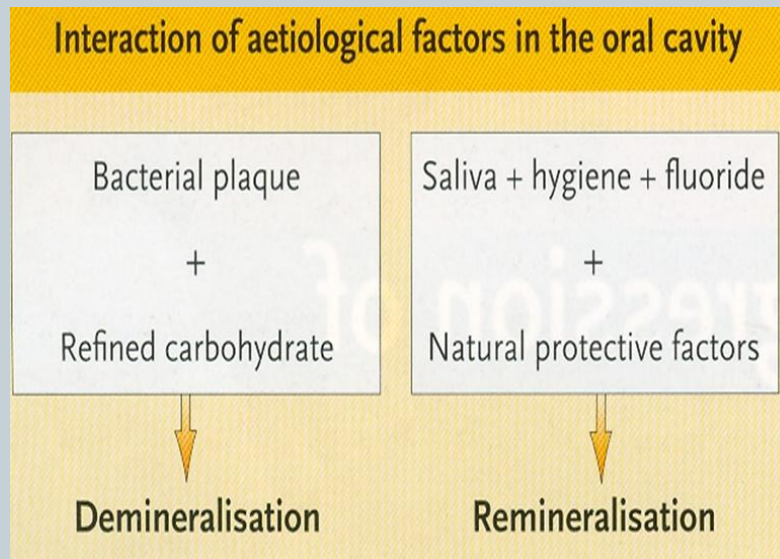


High Risk

1. More than 2 carious lesions in last 3 years
2. No/little fluoride exposure
3. Elevated Streptococcus mutans count
4. Poor oral hygiene
5. Frequent sugar intake
6. Inadequate or poor saliva flow
7. Very irregular dental visits
8. Special health care needs patients

II. Remineralisation of Early carious lesions

- It has been well established that it is possible to arrest and even reverse the mineral loss associated with caries at an early stage, before any cavitation takes place.



Remineralising Agents used for Dental caries prevention



✓ Topical fluoride



✓ Casein Phosphopeptide- Amorphous Calcium Phosphate (CPP-ACP)



✓ Tricalcium phosphate

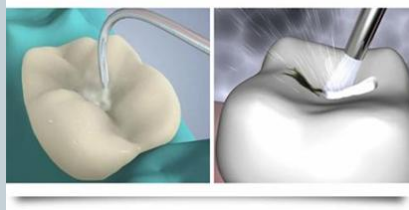


✓ Silver diamine fluoride



III. Minimal intervention procedures for restoration of teeth

1. Air abrasion – Aluminum oxide particles are used



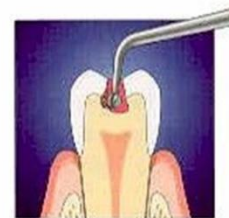
2. Atraumatic restorative treatment(ART)



3. Chemomechanical caries removal (Carisolv)



Apply



Scrape

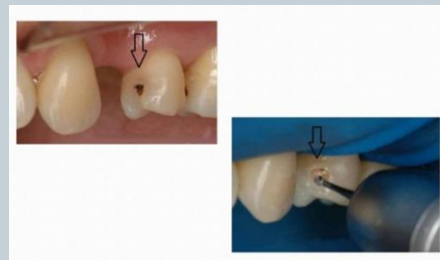
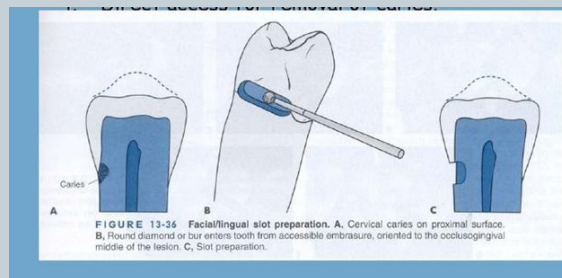


Repair

4. Tunnel preparation



5. Slot preparation



6. Laser cavity preparation



7. Smart polymer burs



IV. Repair/ Replacement restorations?



- Minimal Intervention Dentistry aimed to limit unnecessary removal of healthy tooth structure and repair of defective restorations is one of its strategies.
- Although the repair of resin composite restorations has been investigated extensively and found to be successful
- Defective restorations can be repaired more quickly and with lower operational costs than replacement.



- Repair preserves tooth tissues and is more cost effective, likewise more acceptable to the patient than restoration replacement.



Learning Outcomes



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References



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